

Galois-Informed HRV Analysis

Implementation and Exploratory Assessment

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Status markers

- [B]** *Proved* — algebraically proved.
- [K]** *Confirmed* — numerically verified.
- [S]** *Speculative* — open.
- [L]** *Literature evidence* — external primary source.

Abstract

Docs. 358 and 359 formulate an algebraic hypothesis for HRV frequency analysis: frequency ratios of coupled physiological oscillators should lie on a Galois grid, with the structural tolerance $100\xi \approx 1.33\%$. This document implements the approach as a Python module and assesses it exploratively on two public ECG datasets and on the author's own Polar H10 recordings. The result is negative: the present data do not support the hypothesis, and they cannot test it either, because LF/HF is a power ratio, not a frequency ratio.

Results:

1. **Module [K]**: `galois_hrv_preprocess.py` — Galois projection, HRV band limits, orbit depth, complexity measure $2/(p+q)$.
2. **Null model [K]**: In the grid up to denominator 60, every value between 1 and 4 lies within 1.33 % of a grid element; 39 % of random values lie within 0.06 %. Close grid hits are therefore no evidence on their own.
3. **BIDMC-01 [L]**: intensive-care patient, 88 y, 8 min. No ratio meets the 5-limit criterion; the dataset is unsuitable for the question.
4. **Fantasia [L]**: 5 young and 5 old healthy subjects, about 120 min each, spontaneous breathing. LF peak CV 11–33 %; 5-limit LF/HF 4/10 against a null rate of 21 % (binomial test $p = 0.14$).

5. **Polar H10 [K]**: With validated beat detection, the dominant spectral peak during paced breathing (4–6/min) lies within one frequency bin of the breathing frequency, and the HF peak is its second harmonic. Grid hits of LF/HF occur only at chance frequency.
6. **Structural reason [B]**: Under spontaneous breathing the HF peak is the respiratory frequency and LF/HF the ratio of two independent control loops; with paced breathing below 9/min, respiration itself lies in the LF band.
7. **Next step [S]**: prospective test of frequency coupling during paced breathing (Section 7).

1 Galois HRV preprocessing module

`galois_hrv_preprocess.py` implements (Doc. 359 §5):

Galois grid 901 rational ratios $p/q \leq 4$ with prime factors $\subseteq \{2, 3, 5, 7, 11, 13\}$ up to denominator 60. The prime set is $\{3\} \cup \{p : p \mid 3^k - 1, k \leq 6\}$; a prime $p \neq 3$ belongs to depth k when k is a multiple of $\text{ord}_p(3)$.

Tolerance $100\xi = 1/75 \approx 1.33\%$ with Farey height $Q_c = \pi/\sqrt{6 \cdot 100\xi} \approx 11.1$ (Doc. 343): at this denominator the mean Farey spacing equals twice the tolerance. A grid up to denominator 60 is denser than the tolerance and does not separate.

HRV band limits VLF: $[0, \frac{1}{25}]$ Hz, LF: $[\frac{1}{25}, \frac{3}{20}]$ Hz, HF: $[\frac{3}{20}, \frac{2}{5}]$ Hz. The prime factors lie in $\{2, 3, 5\}$ [B]; the limits are thus compatible with the 5-limit grid, but not determined by it (Section 2).

Galois projection ratio $\mapsto$ nearest p/q in the grid, distance $d = |r/(p/q) - 1|$.

Orbit depth $k = \max \text{ord}_p(3)$ over the prime factors $p \notin \{2, 3\}$ (Docs. 343, 358).

Complexity measure $2/(p+q)$ orders ratios by their simplicity (Doc. 328). It is not a computed Arnold tongue width; that would require the coupling strength of a specific oscillator model.

Pipeline `hrv_galois_pipeline(rr_ms)` — one function call.

Resolution condition. A classification with the tolerance $\delta = 100\xi$ is meaningful only if the frequency resolution satisfies $\Delta f/f \leq \delta$. With $\Delta f = 1/T_w$ for the window length T_w (in Welch's method the segment length, not the recording duration) this gives $T_w \geq 1/(\delta f)$: 12.5 min at 0.10 Hz, 15 min at 0.083 Hz [B]. This is distinct from the minimum separation of two resolvable peaks (about $2/T_w$ for a Hann window) and from the precision of the position of a single peak, which interpolation can make finer than Δf .

Frequency and power ratios. The hypothesis concerns frequency ratios of coupled oscillators. LF/HF is a ratio of integrated band powers; there is no reason to expect it on the grid. In what follows, the projection of LF/HF serves only to show the behaviour of the null model.

2 Band limits and null model

Band limits [K]. The three Task Force limits have prime factors in $\{2, 3, 5\}$ in Hz, mHz, cycles per minute (2.4 = 12/5, 9, 24) and as periods (25 s, 20/3 s, 5/2 s), because the conversion factors 60 and 1000 contain only these primes. In rad/s they are irrational and cannot be classified. The property is not specific either: 8 of the 9 numbers with one significant digit

and a third of all two-decimal values in $(0, 1)$ are 5-limit. Rounded conventions therefore land on the grid by themselves; the band limits are compatible, but not algebraically determined. Frequency ratios, by contrast, are dimensionless and independent of units.

Null model [K]. Probability that a log-uniformly distributed value in $[1, 4]$ lies within the tolerance τ of a grid element (200 000 draws):

τ	Grid up to denominator 60 (901)	Grid up to denominator 11 (133)
1.33 %	100.0 %	90.5 %
0.14 %	73.9 %	18.5 %
0.06 %	39.5 %	7.9 %
0.02 %	14.5 %	2.6 %

With the grid up to denominator 60, the criterion “within 1.33 %” cannot distinguish anything. Even the grid up to the Farey height 11 still covers 91 % at 1.33 %.

3 Negative result I: BIDMC-01

Dataset: BIDMC PPG and Respiration Dataset v1.0.0, record 01 [L] [Pim18]. 88 y, male, MICU, 8 min, 125 Hz PPG, ECG and respiration.

Result: 0/3 frequency ratios meet the 5-limit criterion (Welch and Lomb-Scargle). VLF cannot be resolved in 8 min. DFA $\alpha_1 = 0.50$, $\alpha_2 = 0.45$ — white noise; in healthy subjects ≈ 1.0 .

Why:

1. Too short for VLF.
2. Intensive-care patient: fractal regulation largely lost.
3. HF peak = measured respiratory frequency (0.328 Hz versus 0.336 Hz in the RESP channel).
4. Measurement uncertainty Welch versus Lomb-Scargle $\pm 2\%$, larger than the tolerance of 1.33 %.

4 Negative result II: Fantasia

Dataset: PhysioNet Fantasia Database v1.0.0 [L] [Iye96]. 5 young (f1y01–05, 21–34 y) + 5 old (f2o01–05, 68–85 y), about 120 min rest each, 250 Hz ECG, spontaneous breathing. Reference beat annotations (.ecg), CC-BY 4.0.

Method: wfdb annotations, Welch 10-min windows, zero-padding $\times 4$, parabolic peak interpolation, LF stability over 6×5 -min segments.

Rec	n_{RR}	LF Hz	CV %	LF/HF	p/q	k	5-lim	$2/(p+q)$
f1y01	8707	0.109	11.5	0.438	7/16	6	N	0.087
f1y02	6919	0.085	33.0	0.867	13/15	4	N	0.071
f1y03	7641	0.100	13.7	4.219	4	1	Y	0.400
f1y04	5232	0.078	30.9	1.060	35/33	6	N	0.029
f1y05	6943	0.068	16.5	2.993	3	1	Y	0.500
f2o01	7045	0.067	25.6	1.494	3/2	1	Y	0.400
f2o02	6327	0.052	14.4	0.864	45/52	4	N	0.021
f2o03	6478	0.055	22.0	0.841	21/25	6	N	0.043
f2o04	6849	0.065	32.2	0.719	28/39	6	N	0.030
f2o05	8334	0.049	11.4	1.873	15/8	4	Y	0.087

Statistical tests (Mann-Whitney U): All group differences young versus old n.s. ($p > 0.40$).

Findings:

- LF peak stability: 0/10 records stable ($CV > 1.33\%$).
- 5-limit LF/HF (nearest grid element has prime factors in $\{2, 3, 5\}$): 4/10. The null rate for this criterion is 21 %; one-sided binomial test $p = 0.14$, i.e. not above chance.
- No group difference young/old ($n = 5$ per group).

5 Polar H10: recordings and analysis

Device and protocol: Polar H10 ECG chest strap, firmware 4.2.0, JSONL export via Polar Sensor Recording (Android), the author as the only participant (75 y), 10 September 2026. One spontaneous recording supine (21.9 min, RR intervals computed on the device), six seated recordings with a metronome, 5.0–6.0 min each, 130 Hz ECG. Each click marks a change of respiratory phase; 4, 8, 10 and 12 clicks/min correspond to 2, 4, 5 and 6 breaths/min (0.033 to 0.100 Hz). The two files labelled 4/min are byte-identical; there is one recording at 4/min. Respiration was not measured with a separate sensor.

Analysis [K]: R peaks with NeuroKit2 [L] [Mak21], artefact criterion $|RR/\text{median} - 1| > 0.30$; flagged intervals are removed and the remaining beats stay at their true time stamps. Cubic interpolation to 4 Hz, linear detrending, Hann periodogram over the whole recording ($\Delta f = 1/T \approx 0.003$ Hz), band powers by trapezoidal integration over LF [0.04, 0.15) and HF [0.15, 0.40) Hz.

Beat detection and time axis. The simple energy detector in `polar_h10_atemfrequenz_scan.py` (fixed threshold at the 90th percentile) found 216–285 beats per recording, the validated detector 396–443. In every paced recording 35–47 % of the beats were therefore missing. When the remaining intervals are joined end to end, the time axis shrinks to 49–61 % of the recording duration, and all frequencies shift upwards by a factor of 1.6–2.0. The table is based on the validated detection with the time axis preserved.

Condition	f_R Hz	T min	Beats	RMSSD ms	LF peak	HF peak	LF/HF	nearest p/q
Spontaneous, supine	—	21.9	1517 RR	19.0	0.074	0.225	1.65	33/20 (0.09 %)
2/min, rec. 1	0.033	5.4	431	34.2	0.067	0.200	7.44	—
2/min, rec. 2	0.033	5.1	411	49.6	0.067	0.167	4.24	—
4/min	0.067	5.3	396	67.0	0.067	0.358	4.61	—
5/min, rec. 1	0.083	5.6	431	64.6	0.083	0.167	3.28	105/32 (0.006 %)
5/min, rec. 2	0.083	5.1	400	68.5	0.083	0.167	2.47	121/49 (0.05 %)
6/min	0.100	6.0	443	60.9	0.100	0.201	4.20	—

Frequencies in Hz. —: LF/HF above 4 (outside the grid) or more than 1.33 % from the nearest element. Artefacts: 0–1 each.

Finding 1 — peak at the breathing frequency [K]: At 4, 5 and 6/min the dominant peak lies within one frequency bin of the metronome frequency. Respiration therefore lies in the LF band: the LF peak is the respiratory peak. Where the second harmonic lies above 0.15 Hz (5 and 6/min), it is the HF peak.

Finding 2 — resonance breathing [K]: RMSSD is 61–69 ms at 4–6/min and 34–50 ms at 2/min. This matches the known amplification of HRV by slow breathing near the baroreflex resonance [L] [Leh14, Sev22]. The spontaneous recording was made supine and is not directly comparable with the seated recordings.

Finding 3 — 2/min [S]: The dominant peak lies at 0.067 Hz, twice the intended breathing frequency, with a smaller component at 0.033 Hz. Without a separate respiration signal it

cannot be decided whether this is the second harmonic of very slow breathing or breathing at 4/min because of the click convention.

Finding 4 — grid hits at chance level [K]: The two recordings at 5/min lie close to two *different* grid elements (105/32 and 121/49), although the condition is the same; their LF/HF differs by 28 %. In 17 sliding 300 s windows of the spontaneous recording, LF/HF varies between 0.89 and 7.43 (CV 78 %); of the 13 windows inside the grid range [1, 4], 6 lie within 0.06 % of an element, against a null rate of 39 % (binomial test $p = 0.78$). The spread of the estimate is more than three orders of magnitude larger than the distances to the grid.

Spontaneous recording [K]: mean heart rate 69 /min, RMSSD 19.0 ms, SDNN 48.9 ms. A single missed beat (1741 ms, twice the median) raises RMSSD to 36.3 ms without artefact handling. Across the 300 s windows the LF peak varies between 0.040 and 0.085 Hz (CV 22 %), far above 1.33 %. LF/HF is 1.65 with one window over the whole recording and 2.21 with 300 s segments; a single estimator setting changes the value by a third.

Methodological lessons:

1. Check beat detection against the expected beat count (heart rate $\times$ duration) and a validated detector.
2. Preserve the time axis: remove rejected intervals, keep the remaining beats at their time stamps, never join them end to end.
3. A single artefact can dominate RMSSD; artefact handling is advisable for recordings of any duration.
4. Below 9 breaths/min respiration lies in the LF band; LF/HF then has no interpretation as sympathovagal balance.
5. $n = 1$ and recordings of 5–6 min allow no statistical conclusions; all Polar findings are exploratory.

6 Structural reason and consequence

Finding. [B] Under spontaneous breathing the HF peak is the respiratory frequency (variable, 0.15–0.40 Hz). LF/HF is the ratio of two independent physiological control loops (baroreflex / respiration) and does not converge to a fixed ratio. With paced breathing the metronome sets the frequency; peak and harmonics then stand in trivial integer ratios. In both cases LF/HF is not a test object of the Galois hypothesis.

7 Next step: prospective test

The hypothesis makes a statement about frequency coupling. This can be tested when the breathing frequency is set deliberately relative to the individual baroreflex resonance f_{res} [S]:

- Resonance assessment: paced breathing at 4.5–7/min in steps of 0.5/min, 2 min each [L] [Leh14].
- Four blocks of 21 min (resolution condition at the lowest breathing frequency of about 0.062 Hz) at $\rho = f_R / f_{\text{res}} \in \{2/3, 3/2\}$ (grid) and $\{0.618, 1.618\}$ (golden mean), matched in pairs by breathing rate, in random order.
- Primary outcome: $n:m$ phase-synchronisation index [L] [Tas98] between respiratory phase (respiratory belt) and the HRV component at f_{res} ; contrast grid minus golden mean, paired two-sided t -test.

- Sample size: $d_z = 0.6$, power 0.80 $\Rightarrow n = 24$; with 20 % drop-out, 30 participants.
- Falsification: if coupling is not stronger at grid ratios than at the golden mean, the complexity measure $2/(p+q)$ has no basis in HRV.

A positive result supports the premise but does not yet distinguish the prime set and the Farey height of FFGFT from general mode-locking of coupled oscillators; secondary outcomes serve this purpose (measured frequency ratio against the grid up to denominator 11, peak stability against 1.33 %).

8 Summary

Result	Statement	Status
Module	galois_hrv_preprocess.py works	[K]
Band limits	compatible with the 5-limit, not determined by it	[B]
Null model	close grid hits of LF/HF are no evidence	[K]
BIDMC-01	negative: too short, intensive-care patient	[L]
Fantasia	negative: spontaneous breathing, LF CV 11–33 %, 4/10 at chance level	[L]
Polar H10	peak at the breathing frequency, HF = harmonic, hits at chance	[K]
Reason	LF/HF is a power ratio; respiration determines the peaks	[B]
Next	prospective coupling test, $n = 30$, blocks of 21 min	[S]

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Scripts

2/python/Dok360_Skripte/:
fantasia_galois_analyse.py, galois_hrv_preprocess.py,
01_rpeaks_bidmc01.py, 02_welch_5limit.py, 03_lombscargle_dfa.py,
pruef_galois_nullmodell.py. Polar analysis, null model and all numbers of Sections 2 and 5: SWT_Revision/pruef_swt_revision.py (25/25 assertions). Fantasia data: f1y01-05, f2o01-05 (.ecg/.hea, CC-BY 4.0).

Note on the use of AI assistance

AI assistance (Claude, Anthropic) was used in preparing this document for wording, LaTeX typesetting and analysis scripts. The substantive statements and the responsibility for the text rest with the author.